# Farros Hilmi Syafei

Surabaya, Indonesia · farros.syafei@gmail.com · farroshsyportfolio.vercel.app · github.com/farroshsy · linkedin.com/in/farroshilmisyafei

Full-stack engineer and Informatics Engineering graduate. Ships products end to end: React/Next.js frontends, Python (FastAPI) and TypeScript backends, PostgreSQL, Redis and Kafka, deployed with Docker and Vercel. Built a production document-verification API during an internship and an event-sourced platform of nearly 1,000 source files as a thesis. Works daily with AI coding agents (Claude Code), with tests and CI as the safety net.

## EDUCATION

**Institut Teknologi Sepuluh Nopember (ITS), Surabaya** — Graduated Oct 2026

*B.S. Informatics Engineering · GPA 3.72 / 4.00*

## EXPERIENCE

**Software Engineering Intern — PT Nusantara Baskara Jaya (Nusaraya)** — Dec 2024 – Jun 2025

*Logistics platform · Jakarta* — Python · FastAPI · PaddleOCR · YOLO · Redis · Docker

- Watched how operations staff checked driver and vehicle documents (phone photos retyped by hand), then designed and built a REST API to replace that workflow for onboarding and dispatch.
- Stateless FastAPI service with API-key auth, rate limiting, health checks and metrics. It processes KTP, SIM, STNK and KIR at 95% document-type accuracy and 83–90% field accuracy, and ships as a Docker image (GHCR).
- Designed the API contract around operations: `match / no_match / needs_review` with reason codes, so uncertain cases go to a human. Added perceptual-hash duplicate-upload detection.

**Teaching Assistant, Operating Systems — Dept. of Informatics, ITS** — Feb 2023 – Jun 2023

- Led lab sessions, wrote practicum problems and graded assignments with written feedback.

**Full Stack Web Developer (Bootcamp) — Binar Academy** — Jul 2022 – Jan 2023

- Built MyAirFare, a responsive app to search flights and book round-trip tickets.

## PROJECTS

**HCIE — event-sourced adaptive tutoring platform (thesis)** — FastAPI · PostgreSQL · Kafka · Next.js · React

- Full-stack learning platform: FastAPI backend with a transactional outbox to Kafka feeding CQRS read models, and a Next.js/React frontend using TanStack Query and Recharts dashboards. Runs with Docker Compose and GitHub Actions CI.
- Personalizes for new learners without any interaction history (cold-start AUC 0.6051 vs. BKT 0.5963, on 1.98M interactions from 232k learners). State rebuilds from the event log with bit-identical replay.

**Lyceum — reproducibility index for ML research** — TypeScript · TanStack Start · Bun · PostgreSQL · Typesense · Valkey

- Monorepo with an indexer (arXiv and Papers with Code archive), cron jobs that check papers' repositories and datasets, a React web app with search and paper pages, and an MCP server for AI agents. PostgreSQL behind PgBouncer, with Typesense for search.

**Portfolio with live OCR demo — farroshsyportfolio.vercel.app** — Next.js 16 · React 19 · TypeScript · Tailwind v4 · Vercel

- Content-driven site with a live document-verification demo. A server-side route proxies uploads to the OCR API, keeping the API key server-side, enforcing size, type and document-type limits, and storing nothing.

**dokumen-id — dokumen-id.vercel.app** — Next.js · TypeScript

- Public tool that decodes Indonesian NIK numbers entirely in the browser, with no network requests, so personal data never leaves the device. A Swift package version (DokumenKit) covers iOS.

**rota — self-hostable on-call alerting** — Rust · Docker

- Single-binary service that receives alerts, runs health checks, escalates unacknowledged alerts on a schedule and survives restarts.

## SKILLS

| | |
|---|---|
| **Frontend** | React, Next.js (App Router), TypeScript, TanStack Query/Start, Tailwind CSS, Recharts |
| **Backend** | Python (FastAPI), TypeScript (Bun, Node), Rust, REST API design, event sourcing / CQRS, MCP |
| **Data & Infra** | PostgreSQL, Redis/Valkey, Kafka/Redpanda, Typesense, Docker / Compose, GitHub Actions, Vercel |
| **Also** | Swift/SwiftUI (iOS), PaddleOCR, YOLO, Java, C/C++, SQL; AI-assisted development (Claude Code) |
| **Languages (spoken)** | Indonesian (native), English (fluent) |